

Units & scales

$\hbar c = 197.3\text{ MeV fm}$ $1\text{ eV} = 1.602\times 10^{-19}\text{ J}$
 $\Leftrightarrow 11\,605\text{ K}$ $1\text{ GeV}/c^2 = 1.783\times 10^{-27}\text{ kg}$
 $1\text{ GeV}^{-2} = 0.389\text{ mb} = 3.89\times 10^{-28}\text{ cm}^2$
 $1\text{ GeV}^{-1} = 6.58\times 10^{-25}\text{ s} = 1.97\times 10^{-16}\text{ m}$
 $d \sim \hbar c/E$
 $E_P = \sqrt{\hbar c^5/\overline{G}} = 1.22\times 10^{19}\text{ GeV}$,
 $\ell_P = 1.6\times 10^{-35}\text{ m}$, $M_P^{\text{red}} = 2.4\times 10^{18}\text{ GeV}$
 $F_g/F_e(pp) = \frac{Gm_p^2}{e^2/4\pi\epsilon_0} = 8\times 10^{-37}$
 $Gm_p^2/\hbar c = 5.9\times 10^{-39}$

Relativistic degrees of freedom

$\rho = \frac{\pi^2}{30}g_*T^4$, $g_* = \sum_B g_i + \frac{7}{8}\sum_F g_i$,
 $\rho_i = \frac{g_i T^4}{2\pi^2} \int_0^\infty \frac{x^3 dx}{e^x \mp 1}$
 $\frac{1}{e^x \mp 1} = \frac{1}{e^x - 1} - \frac{2}{e^{2x} - 1} \Rightarrow$
 $I_F = I_B - \frac{2}{16}I_B = \frac{7}{8}I_B$, $I_B = \frac{\pi^4}{15}$
Bosons: $\gamma\,2 + g\,(8\cdot2)\,16 + W^\pm Z\,(3\cdot3)\,9 + H\,1 = 28$
Fermions: $q\,(6\cdot3\cdot2\cdot2)\,72 + \ell^\pm\,12 + \nu\,(3\cdot2)\,6 = 90$
 $g_* = 28 + \frac{7}{8}\cdot90 = \mathbf{106.75}$ check:
 $4\cdot2 + 4 = 12 = 9 + 2 + 1$
 $T \sim \text{MeV}$: $g_* = 2 + \frac{7}{8}(4 + 6) = 10.75$ $e^\pm$
annihilation $g_s : \frac{11}{2} \rightarrow 2$
 $\Rightarrow T_\nu = (\frac{4}{11})^{1/3}T_\gamma = 1.95\text{ K}$; today $g_* = 3.36$,
 $g_{*s} = \frac{43}{11}$

Gauge principle & SM

$\psi \rightarrow e^{i\alpha(x)}\psi$, $D_\mu = \partial_\mu + ieA_\mu$,
 $A_\mu \rightarrow A_\mu - \frac{1}{e}\partial_\mu\alpha$, $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$; $\frac{1}{2}m^2A^2$
not invariant.
 $\mathcal{L} = -\frac{1}{4}F_{\mu\nu}^a F^{a\mu\nu} + i\bar{\psi}\gamma^\mu D_\mu\psi$
 $+ (\bar{\psi}_i y_{ij} \psi_j H + \text{h.c.}) + |D_\mu H|^2 - V(H)$
 $D_\mu = \partial_\mu - ig_s G_\mu^a T^a - igW_\mu^a \frac{\sigma^a}{2} - ig'YB_\mu$
 $G_{\mu\nu}^a = \partial_\mu G_\nu^a - \partial_\nu G_\mu^a + g_s f^{abc}G_\mu^b G_\nu^c$ (3g, 4g
vertices)
 $A = s_W W^3 + c_W B$, $Z = c_W W^3 - s_W B$,
 $e = g s_W = g' c_W$, $s_W^2 = 0.231$

$gT_3W^3 + g'YB = A e(T_3 + Y)$
 $+ Z \frac{g}{c_W}(c_W^2 T_3 - s_W^2 Y)$
 $\gamma: eQ$; $Z: \frac{g}{c_W}(T_3 - s_W^2 Q)$ (using
 $g'/g = s_W/c_W$)
 $g/2 = 1 + \alpha/2\pi + \dots$ strengths @1 GeV:
 $\alpha_s \sim 0.3\text{--}1$, $\frac{1}{137}$, $G_F m_p^2 \approx 10^{-5}$

Representations, charges, anomalies

$SU(2)$: $T^a = \frac{\sigma^a}{2}$, $T_3 = \text{diag}(\frac{1}{2}, -\frac{1}{2})$,
 $T^+ = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$: $d_L \rightarrow u_L$. **3**: rotated by U^* .
 $U(1)$: $\psi \rightarrow e^{iY\theta}\psi$. R-handed field $\equiv$ conjugate
L-handed field, all charges flipped.
field $SU3$ $SU2$ Y mult. $Q = T_3 + Y$
 $Q = (u_L, d_L)$ 3 2 $+\frac{1}{6}$ 6 $+\frac{2}{3}, -\frac{1}{3}$
 u^c $\bar{3}$ 1 $-\frac{2}{3}$ 3 $-\frac{2}{3}$
 d^c $\bar{3}$ 1 $+\frac{1}{3}$ 3 $+\frac{1}{3}$
 $L = (\nu_L, e_L)$ 1 2 $-\frac{1}{2}$ 2 0, -1
 e^c 1 1 $+1$ 1 +1
 $H = (H^+, H^0)$ 1 2 $+\frac{1}{2}$ - +1, 0
 $(T_3 + Y)\langle H \rangle = (-\frac{1}{2} + \frac{1}{2})\langle H \rangle = 0 \Rightarrow Q = T_3 + Y$
unbroken.
Anomaly $\propto \sum_L \text{tr } T_A \{T_B, T_C\}$ ($T(\text{fund}) = \frac{1}{2}$
dropped):
 $[SU3]^2U1$: $2\cdot\frac{1}{6} - \frac{2}{3} + \frac{1}{3} = 0$
 $[SU2]^2U1$: $3\cdot\frac{1}{6} - \frac{1}{2} = 0$
 grav^2U1 : $6\cdot\frac{1}{6} - 3\cdot\frac{2}{3} + 3\cdot\frac{1}{3} - 2\cdot\frac{1}{2} + 1 = 0$
 $U1^3$: $6(\frac{1}{6})^3 + 3(-\frac{2}{3})^3 + 3(\frac{1}{3})^3 + 2(-\frac{1}{2})^3 + 1^3$
 $= \frac{1-32+4-9+36}{36} = 0$
 $SU3^3$: $A(3) = -A(\bar{3}) = 1$: $2 - 1 - 1 = 0$
Witten: $3 + 1 = 4$ doublets
Solve (a, b, c, d, e) = $(YQ, Y_u^c, Y_d^c, Y_L, Y_e^c)$:
 $2a + b + c = 0 \Rightarrow c = -2a - b$;
 $3a + d = 0 \Rightarrow d = -3a$;
 $6a + 3b + 3c + 2d + e = -6a + e = 0 \Rightarrow e = 6a$;
 $(-2a - b)^3 = -(8a^3 + 12a^2b + 6ab^2 + b^3) \Rightarrow$
cubic:
 $6a^3 + 3b^3 - 24a^3 - 36a^2b - 18ab^2 - 3b^3 - 54a^3 + 216a^3$
 $= 144a^3 - 36a^2b - 18ab^2 =$
18a(4a + b)(2a - b) = 0

$b = -4a$: $(\frac{1}{6}, -\frac{2}{3}, \frac{1}{3}, -\frac{1}{2}, 1)$ SM; $b = 2a$: $u \leftrightarrow d$;
 $a = 0$: excluded.
Yukawa $QHuc$: $a + Y_H + b = 0 \Rightarrow Y_H = 3a$;
down $a - 3a + 2a = 0 \checkmark$, e : $-3a - 3a + 6a = 0 \checkmark$;
 $Q = T_3 + \frac{Y}{6a} \Rightarrow Q_u = \frac{2}{3} + \frac{1}{6} = \frac{5}{6}$ for any a .
 $\epsilon_{ij}Q^i H^j = u_L H^0 - d_L H^+ \rightarrow y_u \frac{v}{\sqrt{2}} u_L u^c$;
 $Q_i H^{*i} = u_L H^- + d_L H^{0*} \rightarrow y_d \frac{v}{\sqrt{2}} d_L d^c$;
 $L_i H^{*i} e^c \rightarrow e$ mass only.
Public: $3(\frac{2}{3}) + 3(-\frac{1}{3}) - 1 + 0 = 0$ is $[SU2]^2U1$
($\sum_{\text{doublets}} Q = \sum Y$).

Range & Fermi theory

$(\nabla^2 - m^2)\phi = \delta^3 \Rightarrow V(r) = -\frac{g^2}{4\pi} \frac{e^{-r/\lambda}}{r}$, $\lambda = \frac{\hbar}{mc}$
 $(\Delta E \Delta t \sim \hbar)$
 $\lambda_W = \frac{197.3}{80\,400}\text{ fm} = 2.5\times 10^{-3}\text{ fm} \approx r_p/340$
($r_p = 0.84\text{ fm}$) $\lambda_\pi = 1.4\text{ fm}$
Vertex $\frac{g}{2\sqrt{2}} f \gamma^\alpha (1 - \gamma_5) f' W_\alpha$, propagator
 $\rightarrow \frac{ig_{\alpha\beta}}{m_W^2} \Rightarrow$
 $\frac{G_F}{\sqrt{2}} = \frac{g^2}{8m_W^2} = \frac{g^2}{8g^2 v^2/4} = \frac{1}{2v^2}$
 $v = (\sqrt{2}G_F)^{-1/2} = 246.2\text{ GeV}$
 $G_F = 1.166\times 10^{-5}\text{ GeV}^{-2}$
 $\Gamma_\mu = \frac{G_F^2 m_\nu^5}{192\pi^5} = 3.0\times 10^{-19}\text{ GeV} \Rightarrow \tau_\mu = 2.19\mu\text{s}$
Fermi theory non-unitary at $E \sim G_F^{-1/2} \approx 300$
GeV $G_N = 1/M_P^2$ analogue

Higgs

$V = -\mu^2 H^\dagger H + \lambda(H^\dagger H)^2$, $\langle H \rangle = (0, \frac{v}{\sqrt{2}})$,
 $v^2 = \frac{\mu^2}{\lambda}$, $H^0 = \frac{v+h}{\sqrt{2}}$
 h^2 coeff.: $-\frac{\mu^2}{2} + \frac{6\lambda v^2}{4} = \lambda v^2 \equiv \frac{1}{2}m_H^2 \Rightarrow$
 $m_H^2 = 2\lambda v^2 = 2\mu^2$; $\lambda = 0.13$, $\mu = 88\text{ GeV}$,
 $\frac{v}{m_H} = \frac{1}{\sqrt{2}\lambda} \approx 2$ (with $\langle H^0 \rangle = 174$: 1.39)
 $V \supset \lambda v h^3 + \frac{\lambda}{4} h^4$
 $V(0) - V(v) = \frac{\lambda v^4}{4} \approx (104\text{ GeV})^4 \approx 10^8\text{ GeV}^4$
 $W^a \sigma^a = \begin{pmatrix} W^3 & W^1 - iW^2 \\ W^1 + iW^2 & -W^3 \end{pmatrix}$,
 $(\frac{g}{2}W^a \sigma^a + \frac{g'}{2}B)\begin{pmatrix} 0 \\ v/\sqrt{2} \end{pmatrix} = \frac{v}{2\sqrt{2}}(g(W^1 - iW^2) - gW^3 + g'B)$
 $|D\langle H \rangle|^2 = \frac{v^2}{8}[g^2(W_1^2 + W_2^2) + (gW^3 - g'B)^2]$
 $= \frac{g^2 v^2}{4}W^+ W^- + \frac{1}{2}(\frac{g^2 + g'^2}{4})v^2 Z^2$
 $W^\pm = \frac{W^1 \mp iW^2}{\sqrt{2}}$, $Z = \frac{gW^3 - g'B}{\sqrt{g^2 + g'^2}}$, $A = \frac{g'W^3 + gB}{\sqrt{g^2 + g'^2}}$
massless
 $m_W = \frac{gv}{2}$, $m_Z = \frac{\sqrt{g^2 + g'^2}}{2}v$, $m_W = m_Z c_W$,
 $m_f = \frac{y_f v}{\sqrt{2}}$, $y_t = 0.99$, $y_e = 2.9\times 10^{-6}$; $g = 0.65$,
 $e = 0.31$. Majorana $m_\nu \propto v^2/\Lambda$; $m_p \propto \Lambda_{\text{QCD}}$.
BR(%): $b\bar{b}$ 58, WW^* 21, gg 8, $\tau\tau$ 6, $c\bar{c}$ 3, ZZ^*
2.6, $\gamma\gamma$ 0.23
Metastability: $\lambda < 0$ above 10^{10-11} GeV . d.o.f.:
4 real = 3 eaten + h.
Longitudinal: $k^\mu = (E, 0, 0, k)$, $\varepsilon_\pm = \frac{(0,1,\pm i,0)}{\sqrt{2}}$,
 $\varepsilon_L = \frac{(k,0,0,E)}{m}$, $\varepsilon_L \cdot k = 0$, $\varepsilon_L^2 = -1$,
 $\varepsilon_L \rightarrow \frac{k^\mu}{m} + O(\frac{m}{E}) \Rightarrow$
 $\mathcal{M}(W_L W_L \rightarrow W_L W_L) \sim \frac{E^2}{v^2}$; LQT bound ~ 1
TeV, cancelled by h.

gg → H and the 4th generation

$\mathcal{M} \propto \frac{\alpha_s}{v} \sum_q A_{1/2}(\tau_q)$, $\tau_q = \frac{m_H^2}{4m_q^2}$,
 $A_{1/2} = \frac{2[\tau + (\tau - 1)\arcsin^2 \sqrt{\tau}]}{\tau^2}$
 $\arcsin^2 \sqrt{\tau} = \tau + \frac{\tau^2}{3}$:
 $\tau + (\tau - 1)(\tau + \frac{\tau^2}{3}) = \frac{2}{3}\tau^2 + \frac{1}{3}\tau^3 \Rightarrow A \rightarrow \frac{4}{3}$;
 $\tau_t = 0.13$, $A_t = 1.38$; light: $A \sim \frac{m_g^2}{m_H^2} \ln^2 \rightarrow 0$
LET:
 $\frac{1}{\alpha_s(\mu)} = \frac{\alpha_s(M)}{\alpha_s(M)} + \frac{\beta_0^{(n+1)}}{2\pi} \ln \frac{m_Q}{M} + \frac{\beta_0^{(n)}}{2\pi} \ln \frac{\mu}{m_Q}$,
 $\beta_0 = 11 - \frac{2}{3}n_f$
 $\ln m_Q$ terms: $\frac{-2/3}{2\pi} \ln m_Q = -\frac{\ln m_Q}{3\pi}$;
 $-\frac{1}{16\pi\alpha_s} GG \supset + \frac{\ln m_Q}{48\pi^2} GG$
 $m_Q \rightarrow m_Q(1 + \frac{h}{v})$: $\ln m_Q \rightarrow \ln m_Q + \frac{h}{v}$;
 $G \rightarrow g_s G$:

$\mathcal{L}_{\text{eff}} = \frac{\alpha_s}{12\pi} \frac{h}{v} G_{\mu\nu}^a G^{a\mu\nu}$
per heavy quark, independent of m_Q .

t', b' : $\left(\frac{1.38 + \frac{4}{3} + \frac{4}{3}}{1.38}\right)^2 = \left(\frac{4.05}{1.38}\right)^2 \approx 8.6$; vector-like
quarks evade.

Z line shape & N_ν

$\sigma_{\text{had}}^0 = \frac{12\pi}{m_Z^2} \frac{\Gamma_{e\ell}\Gamma_{\text{had}}}{\Gamma_Z^2}$, $\Gamma_Z = \Gamma_{\text{had}} + 3\Gamma_{\ell\ell} + \Gamma_{\text{inv}}$
 $\Gamma_{f\bar{f}} = N_c \frac{G_F m_Z^3}{6\sqrt{2}\pi} (g_V^2 + g_A^2)$, $g_V = T_3 - 2Qs_W^2$,
 $g_A = T_3$
 ν : $g_V^2 + g_A^2 = \frac{1}{2}$, $\Gamma_{\nu\bar{\nu}} = \frac{G_F m_Z^3}{12\sqrt{2}\pi} = \frac{8.85}{53.3}\text{ GeV}$
 ≈ 166 (167 w/ rad.) MeV
 ℓ : $g_V = -0.038$, $\Gamma_{\ell\ell} = 83.4\text{ MeV}$.
 $\Gamma_{\text{inv}} = 2495.2 - 1744.4 - 3(83.98) = 498.9$
 $N_\nu = 498.9/167.2 = 2.98$ (active, $m < m_Z/2$)
 $N_{\text{eff}}^{\text{SM}} = 3.044$

QCD

$\alpha_s(Q^2) = \frac{12\pi}{(33-2n_f)\ln(Q^2/\Lambda^2)}$, $\Lambda_{\text{QCD}} \approx 0.2\text{--}0.3$
GeV, $\alpha_s(m_Z) = 0.118$; QED $\frac{1}{137} \rightarrow \frac{1}{128}$ at m_Z
 $V(r) = -\frac{4}{3} \frac{\alpha_s \hbar c}{r} + \sigma r$, $\sigma \approx 0.9\text{ GeV/fm}$
 $\approx 1.4\times 10^5\text{ N} \approx 14\text{ t}$
 $3\otimes\bar{3} = 8 \oplus 1$ gluons $m_p = 938\text{ MeV}$ vs
 $2m_u + m_d \approx 9\text{ MeV}$ $T_c \approx 155\text{ MeV}$
 $= 1.8\times 10^{12}\text{ K}$
Clay: quantum YM exists on $\mathbb{R}^4$, mass gap
 $\Delta > 0$ (compact simple G).

Chirality

$P_{L,R} = \frac{1\mp\gamma_5}{2}$; $i\bar{\sigma}^\mu \partial_\mu \psi_L = m\psi_R$,
 $i\sigma^\mu \partial_\mu \psi_R = m\psi_L$; $m=0$:
 $(E + \vec{\sigma}\cdot\vec{p})\psi_L = 0 \Rightarrow \vec{\sigma}\cdot\hat{p}\psi_L = -\psi_L$. L-field:
annihilates f_L , creates f_R ; wrong helicity
 $\propto m/E$.
 $\frac{\Gamma(\pi \rightarrow e\nu)}{\Gamma(\pi \rightarrow \mu\nu)} = \frac{m_e^2}{m_\mu^2} \left(\frac{m_\pi^2 - m_e^2}{m_\pi^2 - m_\mu^2}\right)^2$
 $= 2.34\times 10^{-5} \cdot 5.49 = 1.28\times 10^{-4}$ (meas.
 1.23×10^{-4}); spin $0 \Rightarrow h(\ell^+) = h(\nu) = -\frac{1}{2}$

Neutrinos

$\sigma \sim \frac{G_F^2 E^2}{\pi} (\hbar c)^2 \approx 10^{-44}\text{ cm}^2$ @1 MeV;
 $\lambda_{\text{Pb}} = \frac{1}{n\sigma}$, $n = 7\times 10^{24}\text{ cm}^{-3} \rightarrow 10^{19}\text{ cm} \approx 10\text{ ly}$
Solar: $\frac{6.5}{n\sigma} \times 10^{10}\text{ cm}^{-2}\text{s}^{-1}$ ($\sim 10^{14}/\text{s}$ through
body); relic 336 cm^{-3}
 $P(\nu_\alpha \rightarrow \nu_\beta) = \sin^2 2\theta \sin^2(1.27 \frac{\Delta m^2 [\text{eV}^2] L [\text{km}]}{E [\text{GeV}]})$
 $\Delta m_{21}^2 = 7.4\times 10^{-5}$, $|\Delta m_{31}^2| = 2.5\times 10^{-3}\text{ eV}^2$;
 $\sum m_\nu \geq 0.059$ (NO), 0.10 (IO) eV
Dirac: $\psi_D = (\chi, \xi^\dagger)$, $-m(\chi\xi + \text{h.c.})$, $U(1)$:
 $\chi \rightarrow e^{i\alpha}\chi$, $\xi \rightarrow e^{-i\alpha}\xi$; 4 states
Majorana: $\psi_M = (\chi, \chi^\dagger)$, $-\frac{m}{2}(\chi\chi + \text{h.c.})$,
 $\psi^c = \psi$, 2 states, $\chi\chi \rightarrow e^{2i\alpha}\chi\chi$: $\Delta L = 2$
Weinberg: $\frac{e}{\Lambda}(LH)(LH) \rightarrow m_\nu = \frac{ev^2}{2\Lambda}$,
 $\Lambda \sim 10^{14-15}\text{ GeV}$
Seesaw (ν_L, N^c) : $\begin{pmatrix} 0 & m_D \\ m_D & M \end{pmatrix}$,
 $\lambda_\pm = \frac{M \pm \sqrt{M^2 + 4m_D^2}}{2} \approx M + \frac{m_D^2}{M}$, $-\frac{m_D^2}{M}$;
 $\theta \approx \frac{m_D}{M}$; $m_D = \frac{yv}{\sqrt{2}} \sim 100\text{ GeV}$, $M \sim 10^{14}$
 $\rightarrow 0.1\text{ eV}$
 $0\nu\beta\beta$: $\frac{1}{T_{1/2}} = G_{0\nu}|M_{\text{nucl}}|^2 \left(\frac{m_{\beta\beta}}{m_e}\right)^2$,
 $m_{\beta\beta} = |\sum_i U_{ei}^2 m_i|$; peak at Q ; $T \gtrsim 10^{26}\text{ yr}$,
 $m_{\beta\beta} \lesssim 0.03\text{--}0.1\text{ eV}$

FLRW & Friedmann

$ds^2 = -c^2 dt^2 + a^2 [\frac{dr^2}{1-kr^2} + r^2 d\Omega^2]$
 $1 + z = \frac{a}{a_0} = \frac{\lambda_0}{\lambda_e}$ $T \propto \frac{1}{a}$ $v = H_0 d$
 $G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$
 $H^2 = \frac{8\pi G}{3} \rho - \frac{kc^2}{a^2} + \frac{\Lambda c^2}{3}$
 $\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho + \frac{3p}{c^2}) + \frac{\Lambda c^2}{3}$
 $\dot{\rho} + 3H(\rho + \frac{p}{c^2}) = 0$, $p = w\rho c^2 \Rightarrow \rho \propto a^{-3(1+w)}$;
accelerate iff $w < -\frac{1}{3}$
rad $w = \frac{1}{3}$: $\rho \propto a^{-4}$, $\alpha \propto t^{1/2}$ matter $w = 0$:
 a^{-3} , $t^{2/3}$
 Λ $w = -1$: ρ const, $a \propto e^{Ht}$
 $\rho_c = \frac{3H_0^2}{8\pi G} = 8.5\times 10^{-27}\text{ kg/m}^3 = 5.1\text{ p/m}^3 = 4.8$
GeV/m³

$H^2 = H_0^2(\Omega_r x^4 + \Omega_m x^3 + \Omega_k x^2 + \Omega_\Lambda)$, $x = 1 + z$
 $t_0 = \int_0^\infty \frac{dz}{(1+z)H} = 13.8\text{ Gyr}$, $H_0^{-1} = 14.5\text{ Gyr}$,

$q_0 = \frac{\Omega_m}{2} - \Omega_\Lambda = -0.53$
 Λ =matter: $(1+z)^3 = \frac{\Omega_\Lambda}{\Omega_m} \Rightarrow z = 0.30$; $\ddot{a} = 0$:
 $\Omega_m(1+z)^3 = 2\Omega_\Lambda \Rightarrow z = 0.63$
Observable radius 46 Gly (comoving)

Geometry

$\Omega - 1 = \frac{kc^2}{a^2 H^2} = \pm \left(\frac{c/H}{R_{\text{curv}}} \right)^2, R_{\text{curv}} = \frac{a}{\sqrt{|k|}}:$
 $\Omega > 1$ closed, $= 1$ flat, < 1 open
Newton: $\frac{1}{2}\dot{R}^2 - \frac{GM}{R} = \mathcal{E}, M = \frac{4}{3}\pi\rho R^3 \Rightarrow$
 $\left(\frac{\dot{R}}{R}\right)^2 = \frac{8\pi G}{3}\rho + \frac{2\mathcal{E}}{R^2}, 2\mathcal{E} \leftrightarrow -kc^2r^2$
 $|\Omega - 1| = \frac{|k|c^2}{\dot{a}^2} \propto a^2 \text{ (rad, } \dot{a}^{-2}\propto t), \propto a \text{ (mat, } \dot{a}^{-2}\propto t^{2/3})$
 < 0.002 today $\rightarrow 6 \times 10^{-7}$ ($a_{\text{eq}} = 3 \times 10^{-4}$)
 $\rightarrow 4 \times 10^{-19}$ (BBN) $\rightarrow 10^{-60}$ (t_P)
CMB: $\ell_1 \approx 220/\sqrt{\Omega_{\text{tot}}}$ (spot $\sim 1^\circ$); sound horizon ≈ 150 Mpc comoving

Energy budget

Planck: $\Omega_\Lambda = 0.685, \Omega_c = 0.265$ ($\Omega_c h^2 = 0.120$),
 $\Omega_b = 0.049$ ($\Omega_b h^2 = 0.0224$), $\Omega_\gamma = 5.4 \times 10^{-5}$,
 $h = 0.674, \Omega_k = 0.000 \pm 0.002$
 $\Omega_\nu h^2 = \frac{\sum m_\nu}{93.1 \text{ eV}} (\sum m_\nu = 0.06 \rightarrow 0.14\%)$ stars
 $\approx 0.5\% \approx 10\%$ of baryons

Dark energy

$\rho_{\text{vac}} = \int_0^{k_{\text{max}}} \frac{d^3 k}{(2\pi)^3} \frac{\omega_k}{2} \stackrel{m=0}{=} \frac{1}{4\pi^2} \int_0^{k_{\text{max}}} k^3 dk$
 $= \frac{k_{\text{max}}^4}{16\pi^2} \sim 10^{71} \text{ GeV}^4$ ($k_{\text{max}} = M_P$); dim-reg:
 $\frac{m^4}{64\pi^2} \ln \frac{m^2}{\mu^2}$ per field
 $T_{\mu\nu}^{\text{vac}} = -\rho_{\text{vac}} g_{\mu\nu} \Rightarrow p = -\rho;$
 $\Lambda_{\text{eff}} = \Lambda_{\text{bare}} + 8\pi G \rho_{\text{vac}}$
 $\rho_\Lambda = 2.5 \times 10^{-47} \text{ GeV}^4 = (2.2 \text{ meV})^4;$
 $M_P^4/\rho_\Lambda \sim 10^{120}, \text{EW } 10^{55}$
Piston: $dE = \rho dV = -p dV \Rightarrow p = -\rho;$ source
 $\rho + 3p = -2\rho$
 $S = \int d^4 x \sqrt{-g} \left[-\frac{M^2}{2} R - \frac{1}{2}(\partial\phi)^2 - V \right],$
 $T_{\mu\nu} = \partial_\mu\phi\partial_\nu\phi - g_{\mu\nu}[\frac{1}{2}(\partial\phi)^2 + V]$
 $(\partial\phi)^2 = -\dot{\phi}^2: \rho = T_{00} = \frac{1}{2}\dot{\phi}^2 + V,$
 $p = T_{ii}/a^2 = \frac{1}{2}\dot{\phi}^2 - V$
 $\dot{\rho} + 3H(\rho + p) = 0 \Rightarrow \ddot{\phi} + 3H\dot{\phi} + V' = 0;$
 $w = \frac{\dot{\phi}^2/2 - V}{\dot{\phi}^2/2 + V} \in [-1, 1]$
 $\rho_{\text{DE}}(a) = \rho_0 \exp\left[-3\int_1^a (1+w)\frac{da'}{a'}\right]; \text{CPL}$
 $w = w_0 + w_a(1-a):$
 $\rho \propto a^{-3(1+w_0+w_a)} e^{-3w_a(1-a)}$
 $dL = (1+z) c f \frac{dz}{H} \quad m_\phi^2 = V'' \lesssim H_0^2:$
 $m_\phi \lesssim 1.5 \times 10^{-33} \text{ eV}$
 $V_0 e^{-\lambda\phi/M_P}: w = -1 + \frac{\lambda^2}{3}, \text{ accel. iff } \lambda^2 < 2$
 $V = \Lambda^4(1 + \cos \frac{\phi}{f})$

Thermal history

$H = \frac{1}{2t} = 1.66\sqrt{g_*} \frac{T^2}{M_{\text{Pl}}} \Rightarrow t = \frac{2.4s}{\sqrt{g_*}} \left(\frac{\text{MeV}}{T} \right)^2$
EW crossover 160 GeV 10^{-11} s
sphalerons off 132 GeV
QCD transition 155 MeV 10 μs
 ν dec., n/p freeze 1 MeV 1 s
BBN 0.1–0.03 MeV 3–20 min
equality (z 3400) 0.8 eV 50 kyr
recomb. (z 1090) 0.26 eV 380 kyr
stars / galaxies 0.15 / 0.3 Gyr
accel. / Λ -dom. 7.7 / 10.3 Gyr
 $T_{\text{rec}} = 3000 \text{ K};$ reionization $z \approx 6;$ JWST galaxies $z \approx 14.$
BBN:
 $\frac{n}{p} = e^{-\Delta m/T_f} = e^{-1.293/0.75} \approx \frac{1}{6} \quad \tau_n = 879 \text{ s} \rightarrow \frac{1}{7},$
 $Y_p = \frac{2(n/p)}{1+n/p} = \frac{2/7}{8/7} = 0.25$ (0.245)
 $\eta = 6.1 \times 10^{-10}, n_\gamma = 411 \text{ cm}^{-3}, n_b = 0.25 \text{ m}^{-3},$
 $\text{D/H} = 2.5 \times 10^{-5}, T_0 = 2.7255 \text{ K} = 0.235 \text{ meV},$
 $\delta T/T \sim 10^{-5}$

Inflation

$\frac{d}{dt}(aH)^{-1} = -\frac{\ddot{a}}{a^2}:$ comoving Hubble radius shrinks iff $\ddot{a} > 0$
 $|\Omega - 1| = \left(\frac{c/H}{R_{\text{curv}}} \right)^2 \propto e^{-2Ht}:$
 $N = 60 \Rightarrow e^{-120} \approx 10^{-52}$
 $\ddot{\phi} + 3H\dot{\phi} + V' = 0, H^2 = \frac{1}{3M_P^2}(\frac{1}{2}\dot{\phi}^2 + V)$
 $\epsilon = \frac{M_P^2}{2} \left(\frac{V'}{V} \right)^2, \eta = M_P^2 \frac{V''}{V}, N = \frac{1}{M_P^2} \int \frac{V}{V'} d\phi,$

end at $\epsilon = 1$
 $\mathcal{P}_\zeta = \frac{V}{24\pi^2 \epsilon M_P^4} = 2.1 \times 10^{-9}, n_s - 1 = 2\eta - 6\epsilon,$
 $r = 16\epsilon$
 $V^{1/4} = 1.04 \times 10^{16} (r/0.01)^{1/4} \text{ GeV};$
 $r < 0.036 \Rightarrow V^{1/4} < 1.4 \times 10^{16} \text{ GeV}$
Starobinsky $R + \frac{R^2}{6M^2} / \text{Higgs } \xi |H|^2 R,$
 $\xi \sim 10^{-3-4}; n_s = 1 - \frac{2}{N} = 0.967, r = \frac{12}{N^2} = 0.003$
Data: Planck $0.965 \pm 0.004;$ ACT+DESI
 0.975 ± 0.003 ($\sim 2\sigma$); SPT/Planck 0.965
 $N: \frac{a_e H_e}{a_0 H_0} = \frac{T_0}{T_{\text{reh}}} \cdot \frac{T_{\text{reh}}^2/M_{\text{Pl}}}{H_0} \approx$
 $\frac{(2 \times 10^{-28}) (10^{11} \text{ GeV})}{1.4 \times 10^{-42} \text{ GeV}} \approx 10^{25} \Rightarrow N \gtrsim 58$

Dark matter

$\frac{v^2}{r} = \frac{GM(r)}{r^2} \Rightarrow v = \sqrt{\frac{GM(r)}{r}};$ flat $\Rightarrow M \propto r,$
 $\rho \propto r^{-2}; \text{MW } 230 \text{ km/s } @ 8 \text{ kpc} \rightarrow 10^{11} M_\odot$
Virial $M \sim R\sigma_v^2/G \quad \text{NFW } \rho = \frac{\rho_s}{(r/r_s)(1+r/r_s)^2}$
 $\delta \propto a$ (matter era)
 $\rho_\odot = 0.3\text{--}0.4 \text{ GeV/cm}^3, v \approx 230 \text{ km/s}; 100 \text{ GeV}:$
4 per litre, $10^5 \text{ cm}^{-2} \text{s}^{-1}$ MOND
 $a_0 = 1.2 \times 10^{-10} \text{ m/s}^2$
Freeze-out: $\dot{n} + 3Hn = -\langle\sigma v\rangle(n^2 - n_{\text{eq}}^2),$
 $n_{\text{eq}} \propto (mT)^{3/2} e^{-m/T}, n\langle\sigma v\rangle = H @$
 $x_f = \frac{m}{T} \approx 20\text{--}25; \rho_0 \propto x_f \frac{T_0^3}{M_{\text{Pl}}\langle\sigma v\rangle}$ (mass-indep.)
 $\Omega_\chi h^2 \approx \frac{3 \times 10^{-27} \text{ cm}^3 \text{s}^{-1}}{\langle\sigma v\rangle}$
 $0.12 \Rightarrow \langle\sigma v\rangle = 2.2 \times 10^{-26} \text{ cm}^3/\text{s};$
 $1 \text{ GeV}^{-2} \cdot c = 1.17 \times 10^{-17} \text{ cm}^3/\text{s}$
 $\Rightarrow 2 \times 10^{-9} \text{ GeV}^{-2} \sim \frac{a^2}{m^2}, \alpha = 0.01 \rightarrow m \approx 200 \text{ GeV}$
Freeze-in: $\lambda \sim 10^{-11-12}, \Omega \propto \lambda^2$ masses 10^{-22}
eV $\sim \sim 30 M_\odot;$ fermion $\gtrsim \text{keV}$
Direct: $E_R = \frac{\mu^2 v^2}{m_N} (1 - \cos\theta), \mu = \frac{m_\chi m_N}{m_\chi + m_N},$
 $E_R^{\text{max}} = \frac{2\mu^2 v^2}{m_N}; 100 \text{ GeV on Xe } (m_N = 122): 30 \text{ keV}$
 $\frac{dR}{dE_R} = \frac{\rho_\chi}{m_\chi m_N} \int_{v_{\text{min}}}^{v_f} v f(v) \frac{d\sigma}{dE_R} d^3 v, \text{SI } \propto A^2$
Indirect: $\frac{d\Phi_\gamma}{dE} = \frac{4\pi v}{8\pi m_\chi^2} \frac{dN_\gamma}{dE} \int_{\Delta\Omega} \int_{\text{los}} \rho_\chi^2 dl d\Omega$
 $\text{BR}(H \rightarrow \text{inv}) \lesssim 10\%$

Sphalerons & the 28/79

$\partial_\mu J_B^\mu = \partial_\mu J_L^\mu = N_g \frac{g^2}{32\pi^2} W_{\mu\nu} \tilde{W}^{a\mu\nu} - (g' \text{ term});$
per gen. $B: 3 \cdot \frac{1}{3}, L: 1$
 $\Delta B = \Delta L = N_g \Delta N_{\text{CS}}, N_{\text{CS}} =$
 $\frac{g^2}{32\pi^2} \int d^3 x \epsilon^{ijk} (W_i^a \partial_j W_k^a + \frac{g}{3} \epsilon^{abc} W_i^a W_j^b W_k^c),$
 $\pi_3(S^3) = \mathbb{Z}$
 $\Delta N_{\text{CS}} = 1: u_L d_L d_L \nu_e + c_L s_L s_L \nu_\mu + t_L b_L b_L \nu_\tau$
 $(9q+3\ell), \Delta(B-L) = 0$
 $E_{\text{sph}} = \frac{4\pi v}{g} B(\frac{\lambda}{g^2}), B = 1.5\text{--}2.7 \Rightarrow \approx 9 \text{ TeV};$ at
 $N_{\text{CS}} = \frac{1}{2}, \text{size } \sim 1/m_W$
 $T = 0: e^{-4\pi/\alpha_W} = e^{-16\pi^2/g^2} \approx 10^{-160};$ broken:
 $e^{-E_{\text{sph}}(T)/T}; \text{symm.}: \Gamma \approx 20\alpha_W^5 T^4$
 $n - \bar{n} = \frac{gT^2}{6} \text{ (F)}, \times 2 \text{ (B)}. \mu_u = \mu_q + \mu_H,$
 $\mu_d = \mu_q - \mu_H, \mu_e = \mu_\ell - \mu_H; \text{sph.}: \mu_\ell = -3\mu_q$
 $Y = 0: N_g[\mu_q + 2\mu_u - \mu_d - \mu_\ell - \mu_e] + 2m\mu_H =$
 $N_g(2\mu_q - 2\mu_\ell + 4\mu_H) + 2m\mu_H = 0$
 $\Rightarrow \mu_H = -\frac{4N_g\mu_q}{2N_g+m}$
 $B = \frac{N_g}{3}(6\mu_q + 3\mu_u + 3\mu_d) = 4N_g\mu_q,$
 $L = N_g(3\mu_\ell - \mu_H) = N_g\mu_q \frac{-14N_g-9m}{2N_g+m}$
 $\frac{B}{B-L} = \frac{8N_g+4m}{22N_g+13m} = \frac{28}{79} \quad (N_g=3, m=1)$

Leptogenesis

$\mathcal{L} \supset -y_{\alpha i} \bar{\ell}_\alpha \tilde{H} N_i - \frac{M_i}{2} \bar{N}_i N_i; \Gamma_{N_1} = \frac{(y^\dagger y)_{11} M_1}{8\pi};$
 $N_1 \rightarrow \ell H (+1), \bar{\ell} H^* (-1)$
 $A = ae^{i\phi_a} e^{i\delta a} + be^{i\phi_b} e^{i\delta b}, \bar{A}: \phi \rightarrow -\phi \Rightarrow$
 $|A|^2 - |\bar{A}|^2 = -4ab \sin\Phi \sin\Delta$
 $\epsilon_1 \approx -\frac{3}{16\pi} \frac{1}{(y^\dagger y)_{11}} \sum_j \text{Im}[(y^\dagger y)_{1j}^*] \frac{M_j}{M_j}; \text{DI:}$
 $|\epsilon_1| \leq \frac{3}{16\pi} \frac{M_1(m_3-m_1)}{v^2} \approx 10^{-6} \frac{M_1}{10^{10} \text{ GeV}} \quad (v=174)$
 $K = \frac{\Gamma_{N_1}}{H(M_1)} = \frac{\tilde{m}_1}{m_*}, \tilde{m}_1 = \frac{(y^\dagger y)_{11} v^2}{M_1},$
 $m_* = 1.1 \times 10^{-3} \text{ eV} \Rightarrow K \sim 10\text{--}50$
 $\frac{dN_N}{dz} = -D(N_N - N_N^{\text{eq}}),$
 $\frac{dN_{B-L}}{dz} = -\epsilon_1 D(N_N - N_N^{\text{eq}}) - WN_{B-L}, z = \frac{M_1}{T}$
 $\eta_B = \frac{28}{79} \cdot \frac{3}{4} \cdot \frac{86}{2387} \epsilon_1 \kappa_f = 0.96 \times 10^{-2} \epsilon_1 \kappa_f \quad (g_* \text{ :}$
 $108.5 \rightarrow \frac{41}{13}) \Rightarrow M_1 \gtrsim 10^9 \text{ GeV}; B = -\frac{28}{79} \Delta L$

Network: decays; inverse decays; $\Delta L=2$
 $\ell H \leftrightarrow \ell H^*, \bar{\ell} \bar{H} \leftrightarrow H^* H^* \text{ (subtract on-shell } N);$
 $\Delta L=1 \ N \ell \leftrightarrow \bar{q} t, N t \leftrightarrow \bar{\ell} q$

EW baryogenesis

$V(\phi, T) = D(T^2 - T_0^2)\phi^2 - ET\phi^3 + \frac{\lambda}{4}\phi^4;$ from
 $-\frac{T}{12\pi} \sum_b n_b m_b^3, n_W=6, n_Z=3:$
 $E = \frac{2m_W^3 + m_Z^3}{4\pi v^3} = 0.0096$
 $D = \frac{2m_W^2 + m_Z^2 + 2m_t^2}{8v^2} = 0.17$
 $T_c: V = \frac{\lambda}{4}\phi^2(\phi - \phi_c)^2 = \frac{\lambda}{4}(\phi^4 - 2\phi_c\phi^3 + \phi_c^2\phi^2)$
 $\Rightarrow ET_c = \frac{\lambda\phi_c}{2}, \frac{\phi_c}{T_c} = \frac{2E}{\lambda} = 0.15$

Washout:
 $T e^{-E_{\text{sph}}/T} < \frac{T^2}{M_{\text{Pl}}} \Rightarrow \frac{E_{\text{sph}}}{T} \gtrsim \ln \frac{M_{\text{Pl}}}{T} \approx 40,$
 $E_{\text{sph}} \approx 37\phi \Rightarrow \frac{\phi_c}{T_c} \gtrsim 1$
Nucleation $\Gamma \propto T^4 e^{-S_3/T}, S_3/T \approx 140$
Singlet:
 $+\frac{1}{2}\mu_S^2 S^2 + \frac{1}{4}\lambda_S S^4 + \frac{1}{2}\lambda_{HS} |H|^2 S^2 (+a_1 |H|^2 S);$
 $(0, w) \rightarrow (v, 0)$
 $m_t(z) = \frac{y_t}{\sqrt{2}} \phi(z) (1 + i \frac{S(z)}{\Lambda}) = |m_t| e^{i\theta(z)};$
 $F_z = -\frac{(m|^2)'}{2\omega} \pm s \frac{(m|^2\theta')'}{2\omega\omega_z}$
 $\partial_t n_B \propto -N_g \frac{\Gamma_{\text{WS}}}{T^3} n_L(z) - \mathcal{R} n_B$
 $|de| < 4.1 \times 10^{-30} \text{ e cm}$
GW: $f \approx 1.9 \times 10^{-5} \text{ Hz } \frac{\beta}{v_w} \frac{1}{H} \frac{T}{100 \text{ GeV}} \left(\frac{g_*}{100} \right)^{1/6} \sim$
mHz (LISA)

Particle data

u 2.16 MeV c 1.273 GeV t 172.6 GeV
 d 4.70 MeV s 93.5 MeV b 4.183 GeV
 e 0.511 MeV μ 105.66 MeV τ 1776.9 MeV
 $\tau_\mu = 2.197 \mu\text{s}$ ($c\tau = 659 \text{ m}$), $\tau_\tau = 290 \text{ fs},$
 $\tau_n = 878 \text{ s}, \tau_t \approx 5 \times 10^{-25} \text{ s}$
 $\Gamma_Z = 2.495, \Gamma_W = 2.085, \Gamma_t = 1.42 \text{ GeV},$
 $\Gamma_H = 4.1 \text{ MeV}$
SM parameters: 3 gauge + 6 m_q + 3 m_ℓ + 4
CKM + 2 Higgs + $\theta_{\text{QCD}} = 19$ (+7/9 with ν)
1 pc = $3.086 \times 10^{16} \text{ m} = 3.26 \text{ ly}; 1 \text{ yr}$
 $= 3.156 \times 10^7 \text{ s}; M_\odot = 1.99 \times 10^{30} \text{ kg} = 1.1 \times 10^{57} \text{ GeV};$
 $H_0 = 2.18 \times 10^{-18} \text{ s}^{-1} = 1.44 \times 10^{-42} \text{ GeV}$

Public-question formulas

GPS ($r = 26571 \text{ km}$):
 $\frac{GM_E}{c^2} \left(\frac{1}{R_E} - \frac{1}{r} \right) = 5.3 \times 10^{-10} \rightarrow +45.7 \mu\text{s/d};$
 $-\frac{v^2}{2c^2} = -\frac{GM_E}{2rc^2} = -8.4 \times 10^{-11} \rightarrow -7.2 \mu\text{s/d};$
net +38.5 $\mu\text{s/d}$ $\times c \approx 11 \text{ km/d}$
Cosmic ray vs LHC:
 $\sqrt{s} = \sqrt{2m_p E} = \sqrt{2 \cdot 0.938 \cdot 10^{11}} \text{ GeV} = 430 \text{ TeV}$ vs 13.6 TeV
Muon: $E = 3 \text{ GeV} \Rightarrow \gamma = 28, \gamma c\tau = 19 \text{ km}$
Antimatter: $2g \cdot c^2 = 1.8 \times 10^{14} \text{ J} = 43 \text{ kt TNT}$
 $r_s = \frac{2GM}{c^2} = 2.95 \text{ km } \frac{M}{M_\odot};$
 $T_H = \frac{\hbar c^3}{8\pi GM k_B} = 6.2 \times 10^{-8} \text{ K } \frac{M_\odot}{M};$
 $t_{\text{evap}} \sim 10^{67} \text{ yr } \left(\frac{M}{M_\odot} \right)^3$

Numbers & news

$m_{W,Z,H,t}$ 80.37, 91.19, 125.2, 172.6 GeV
 α, α_s, s_W^2 1/137, 0.118, 0.231
 H_0 Planck $67.4 \pm 0.5;$ SH0ES $73.2 \pm 0.9;$ CCHP 70.4
age, T_0, z_{rec} 13.8 Gyr, 2.7255 K, 1090
 $\eta, Y_p, N_{\text{eff}}$ $6.1 \times 10^{-10}, 0.245, 3.044$
DESI DR2 DDE 3.1 σ (+CMB), 2.8–4.2 σ (+SN)
 $w_0 \approx -0.7$ to $-0.85, w_a \approx -0.6$ to -1.1
 $\sum m_\nu$ $< 0.064 \text{ eV}$ (ΛCDM), < 0.16 ($w_0 w_a$)
KATRIN $m_\nu < 0.45 \text{ eV}$ (90%)
LZ (1 Sep 2026) 1 event, 248 keV NR, 2.6 σ global
XENONnT null; near ν fog
(Aug 2026)
 $p \rightarrow e^+ \pi^0$ $> 2.4 \times 10^{34} \text{ yr}$
FCC-ee preferred; decision 2028
 σ $2.6\sigma \approx 0.5\%; 5\sigma = 1/3.5M$